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(54) **SYSTEM AND APPARATUS FOR
REACTIONS INCLUDING A LIQUID
TRANSFER DEVICE WITH AN
ASYMMETRICAL CROSS-SECTION**

division of application No. 15/141,190, filed on Apr. 28, 2016, now Pat. No. 10,040,061, which is a continuation of application No. 13/242,999, filed on Sep. 23, 2011, now Pat. No. 9,352,312.

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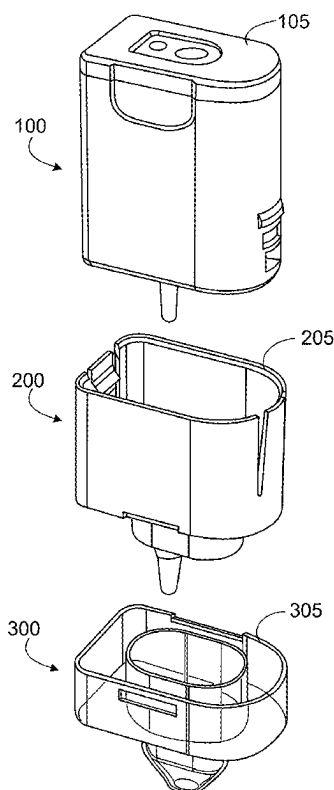
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(57) **ABSTRACT**

This disclosure provides systems, apparatuses, and methods for liquid transfer and performing reactions. In one aspect, a system includes a liquid transfer device having a housing having a pipette tip and a plunger assembly; and a reaction chamber, wherein the housing of the liquid transfer device is configured to sealably engage with the reaction chamber. In another aspect, a liquid transfer device including a housing having a pipette tip; and a plunger assembly disposed within the housing and the pipette tip, wherein a portion of the plunger assembly is configured to engage a fluid reservoir such that the plunger assembly remains stationary relative to the fluid reservoir and the housing moves relative to the plunger assembly.



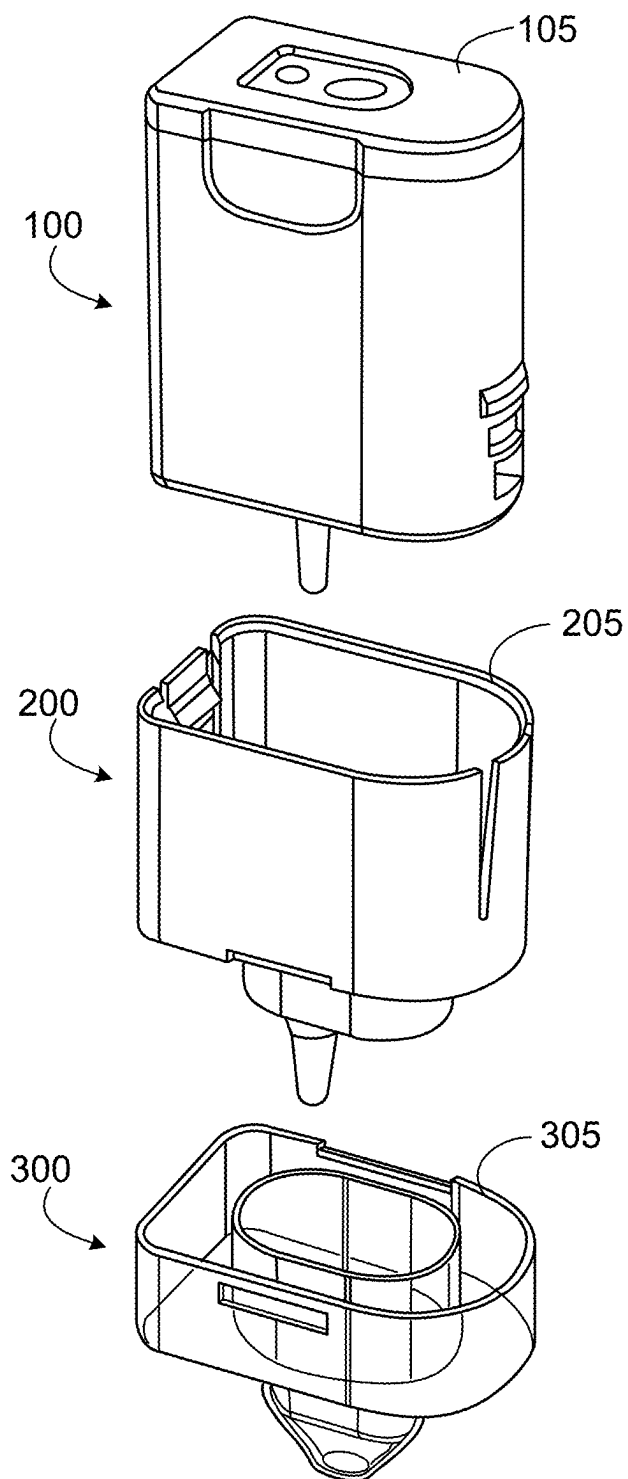


FIG. 1

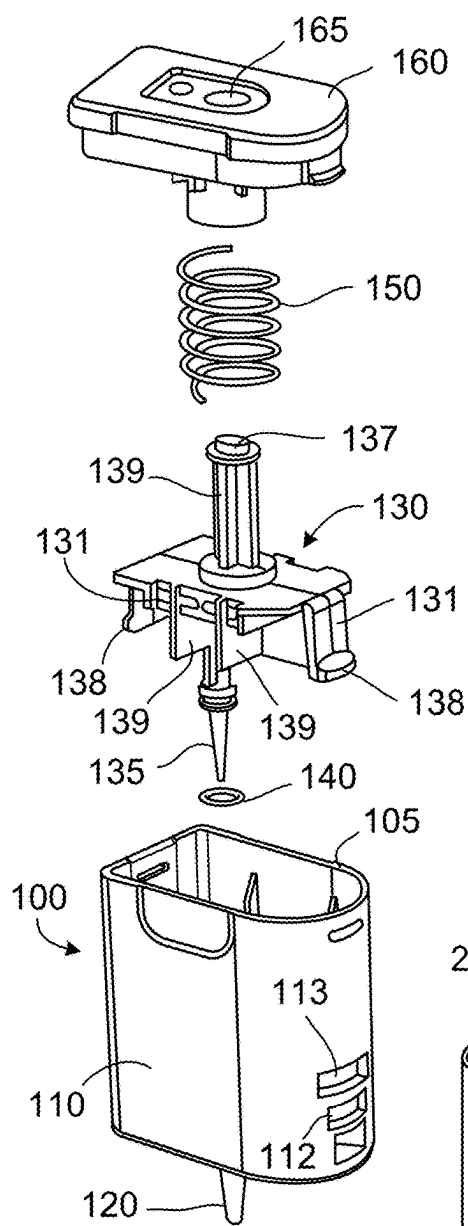


FIG. 2A

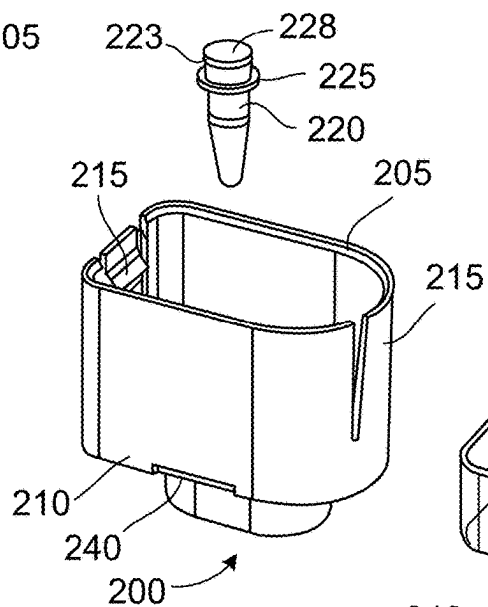


FIG. 2B

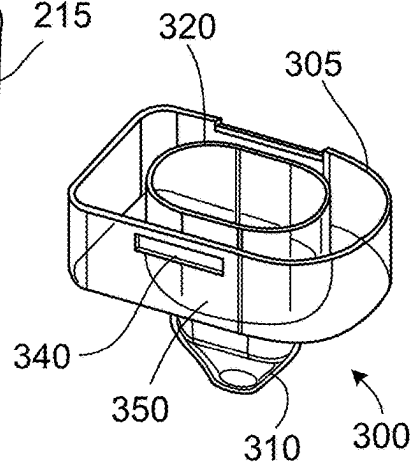


FIG. 2C

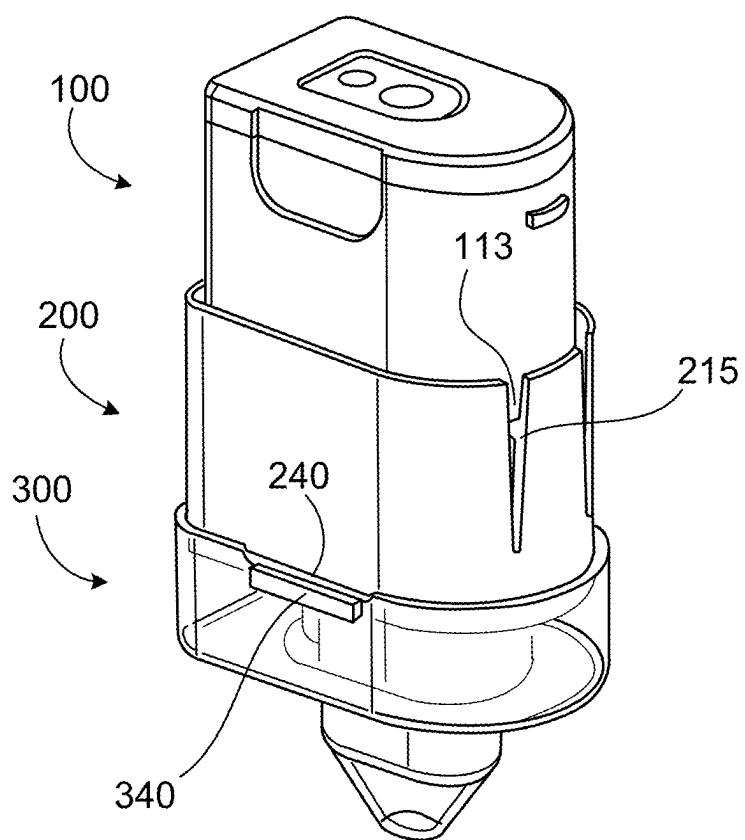
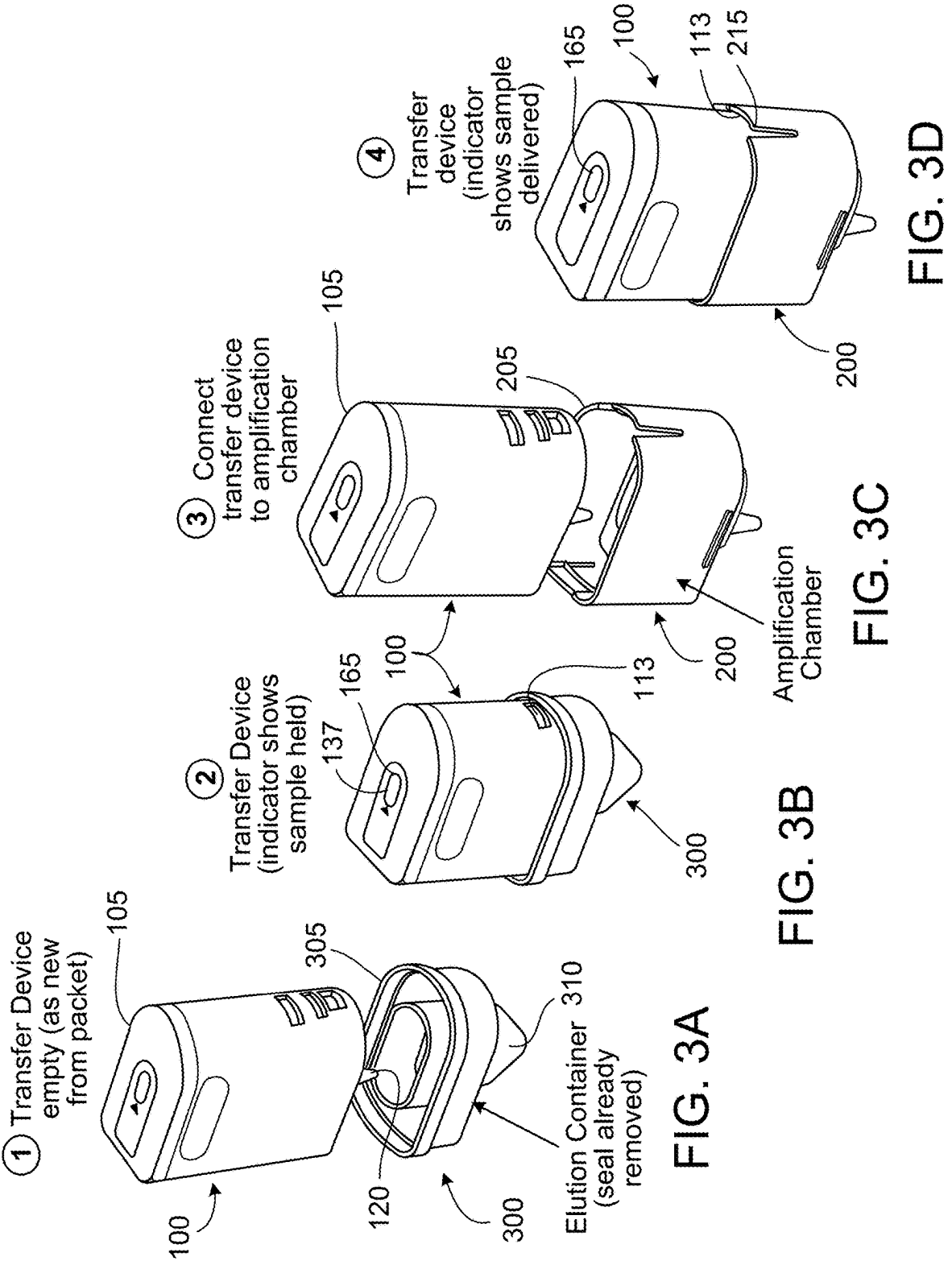


FIG. 2D



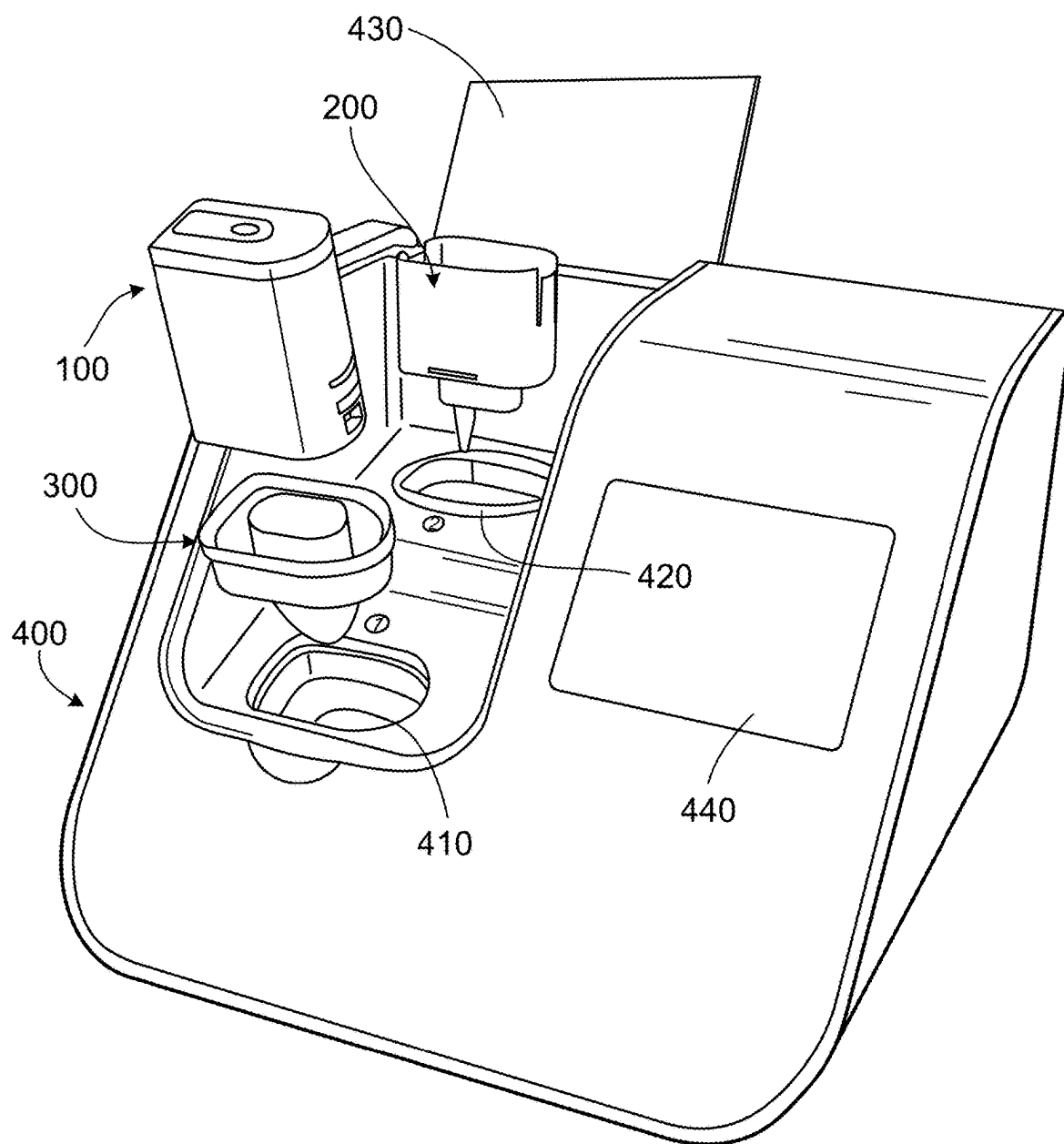


FIG. 4

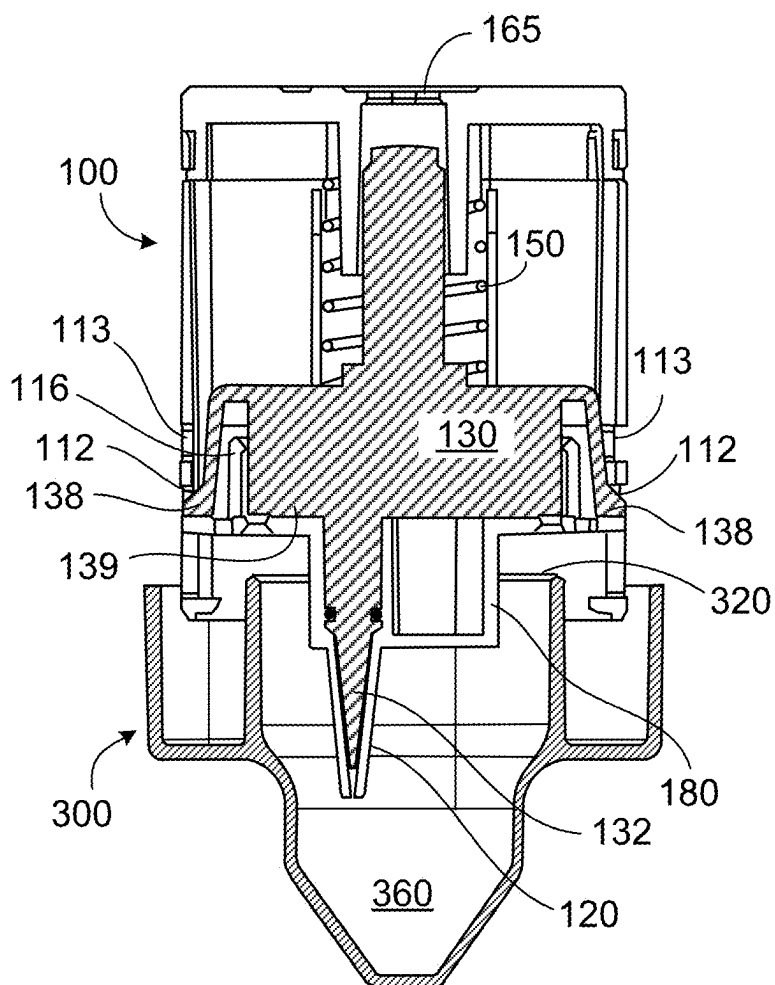


FIG. 5A

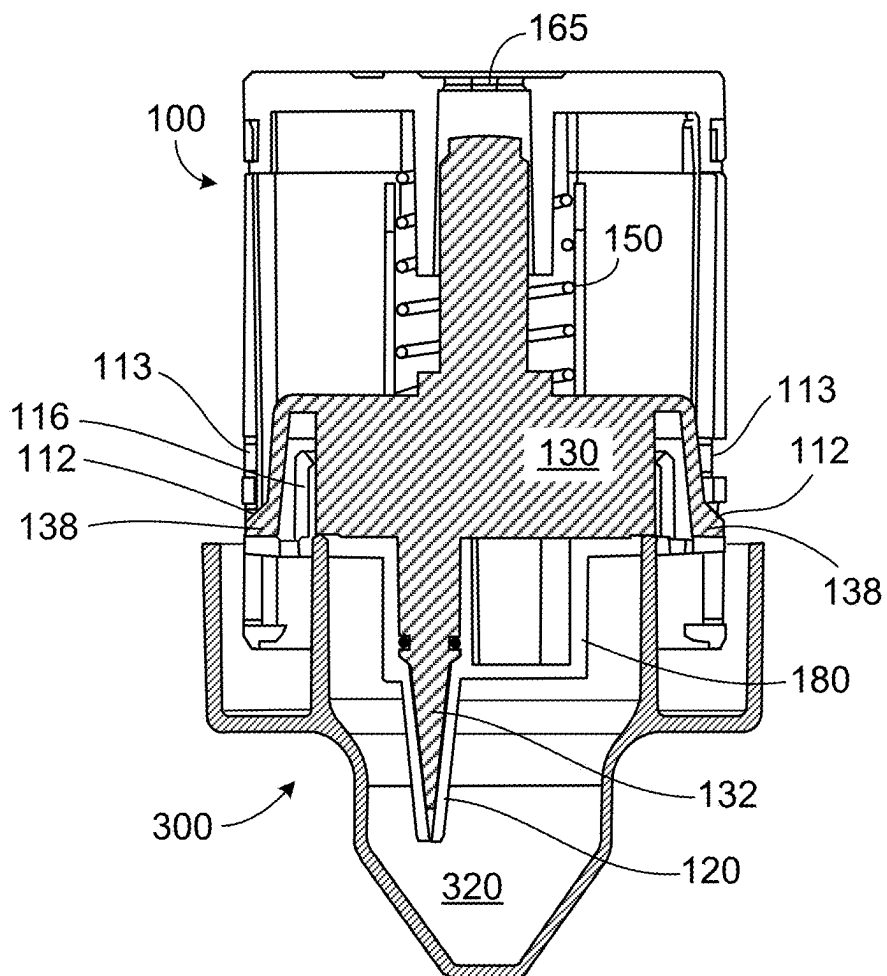


FIG. 5B

FIG. 5C

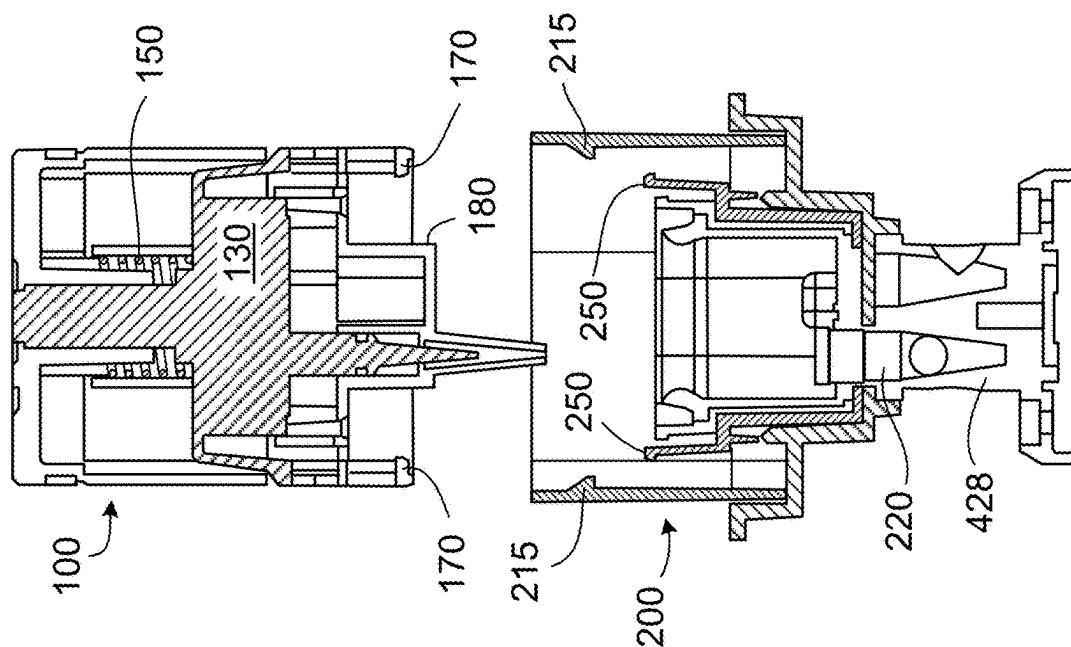


FIG. 6A

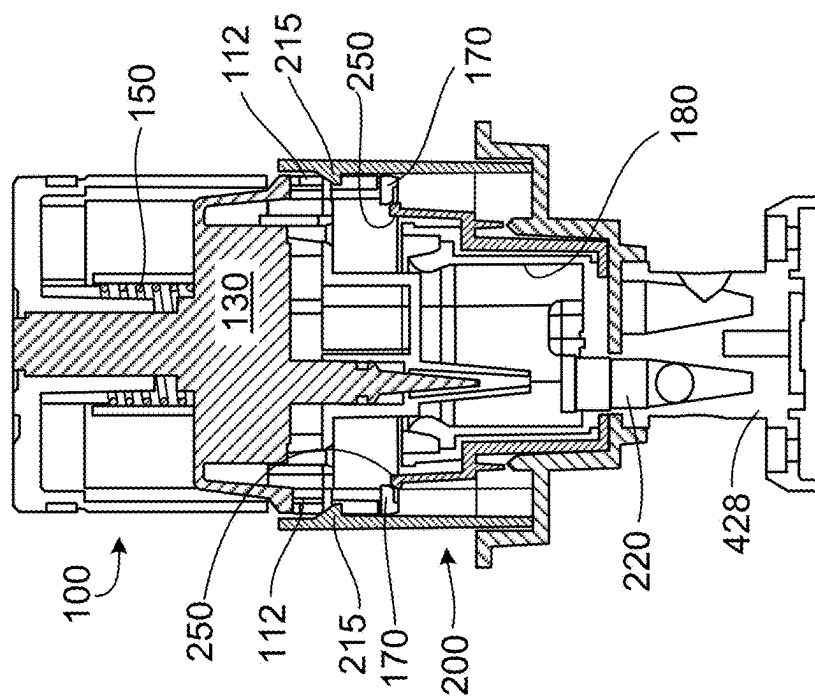


FIG. 6B

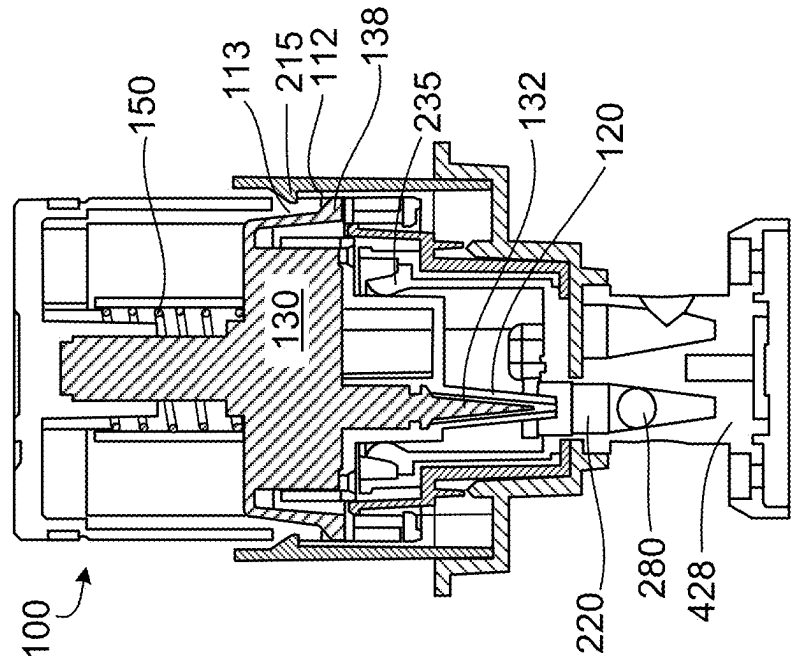


FIG. 6D

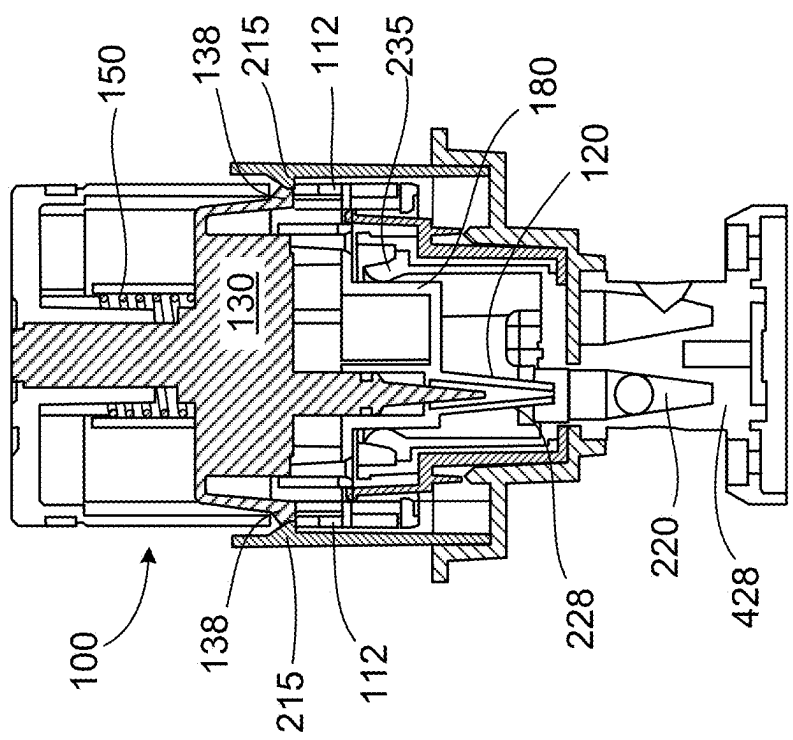
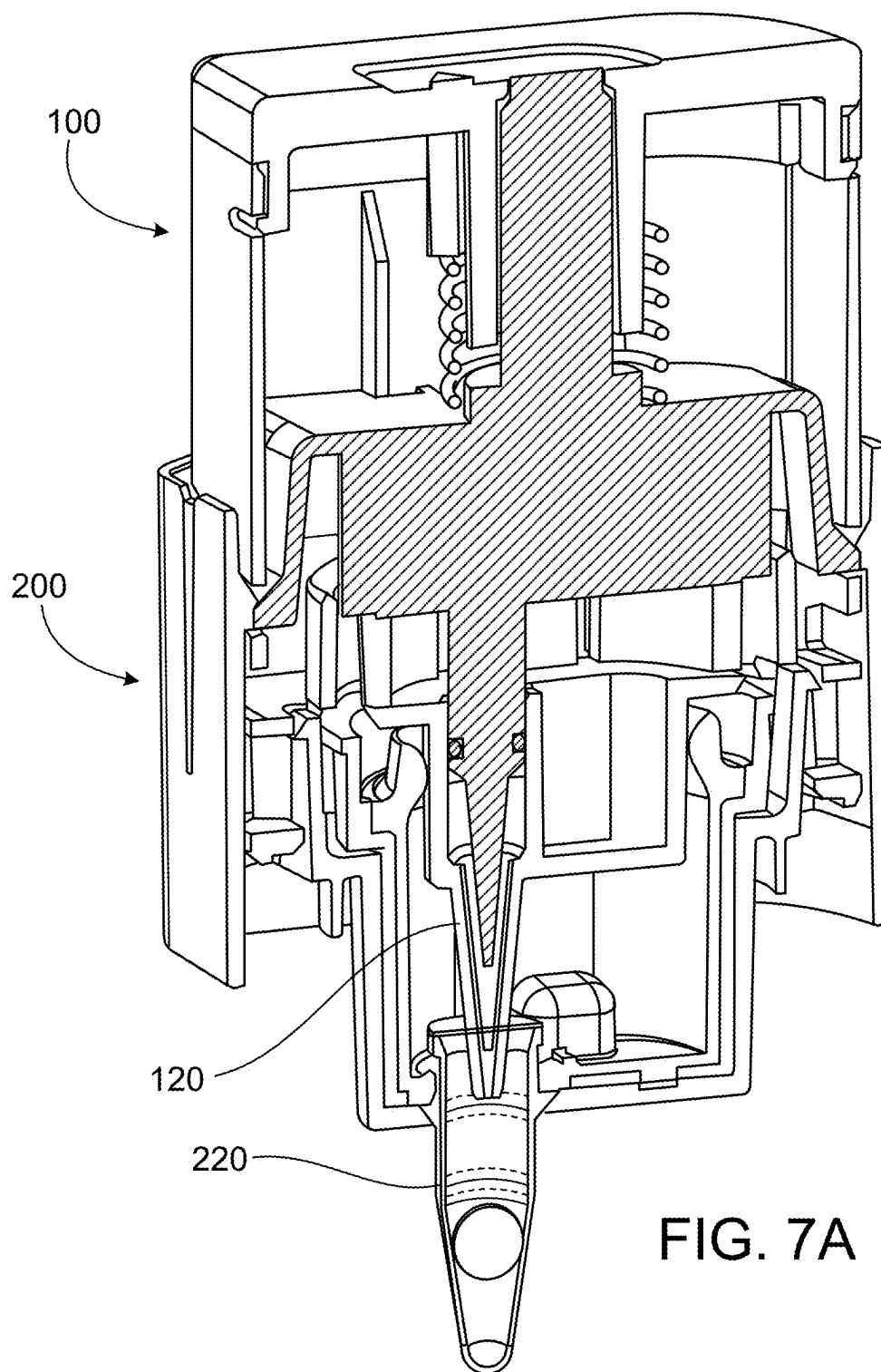


FIG. 6C



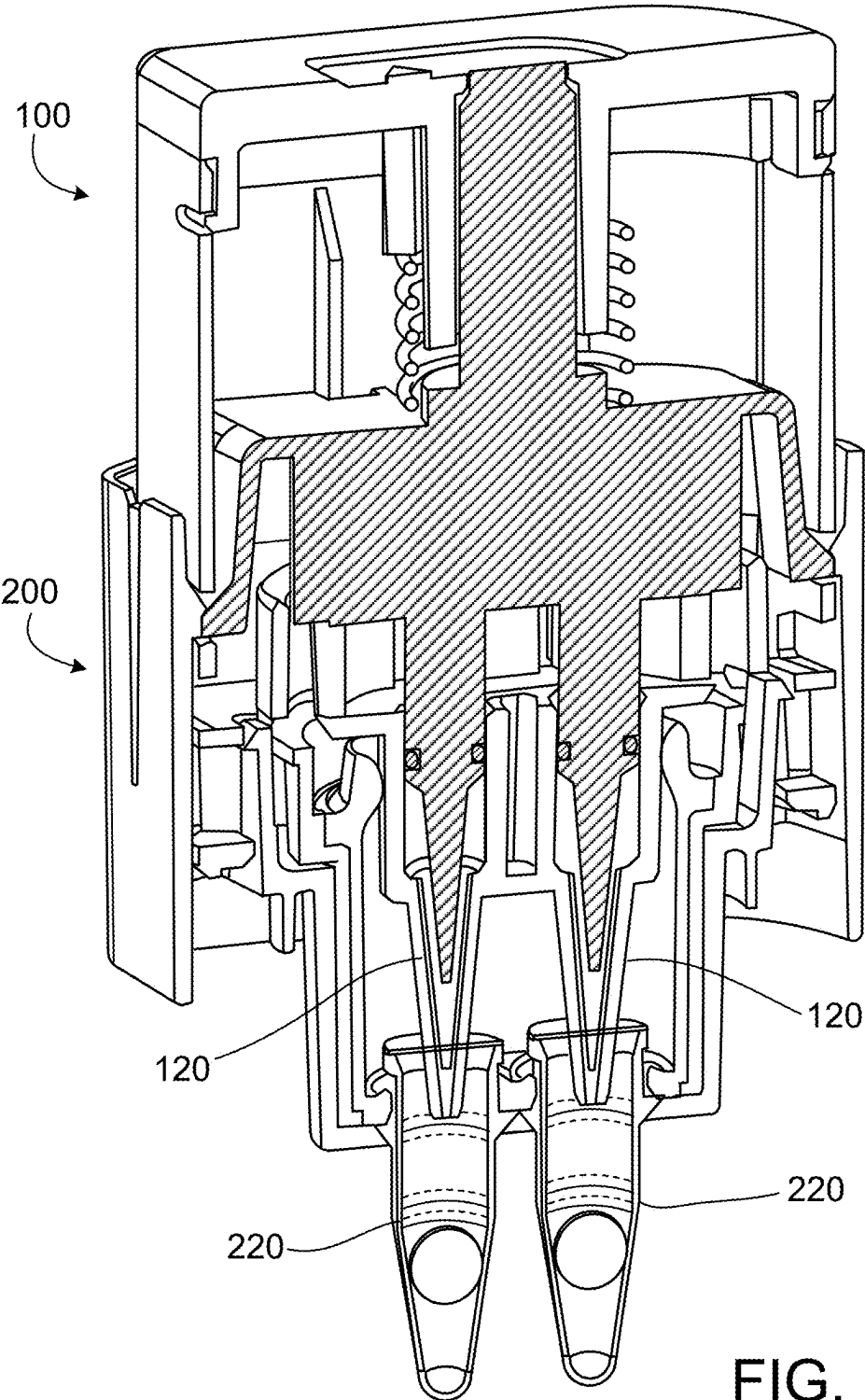


FIG. 7B

**SYSTEM AND APPARATUS FOR
REACTIONS INCLUDING A LIQUID
TRANSFER DEVICE WITH AN
ASYMMETRICAL CROSS-SECTION**

**CROSS-REFERENCE TO RELATED
APPLICATION**

[0001] This application is a continuation of U.S. patent application Ser. No. 17/238,841, filed Apr. 23, 2021, now allowed, which is a continuation of U.S. patent application Ser. No. 16/057,209, filed Aug. 7, 2018, now U.S. Pat. No. 11,033,894, issued Jun. 15, 2021, which is a divisional of U.S. patent application Ser. No. 15/141,190, filed Apr. 28, 2016, now U.S. Pat. No. 10,040,061, issued Aug. 7, 2018, which is a continuation and claims priority to U.S. patent application Ser. No. 13/242,999, filed Sep. 23, 2011, now U.S. Pat. No. 9,352,312, issued May 31, 2016, the entire contents of each of which are incorporated by reference.

TECHNICAL FIELD

[0002] This invention relates to systems and apparatuses for liquid transfer and carrying out reactions.

BACKGROUND

[0003] Many diagnostic tests that involve biological reactions are required to be performed in laboratories by skilled technicians and/or complex equipment. Such laboratories may be the subject of government regulation. The costs of compliance with such regulations can increase the costs of diagnostic tests to patients and health care payers and exclude such tests from point-of-care facilities. There is a need for systems for performing diagnostic tests involving biological reactions that can be used without extensive training at the point of care.

SUMMARY

[0004] The present disclosure provides systems, apparatuses and methods for transfer of liquids and processing of reactions, e.g., in diagnostic tests.

[0005] In one aspect, the disclosure features a system that includes a liquid transfer device that includes a housing having a pipette tip and a plunger assembly; and a reaction chamber, wherein the housing of the liquid transfer device is configured to sealably engage with the reaction chamber. In some embodiments, the housing of the liquid transfer device can include a seal component configured to sealably engage with the reaction chamber. In some embodiments, the reaction chamber can include a seal component configured to sealably engage with the liquid transfer device. The systems can further include a fluid reservoir, and the reaction chamber can optionally be configured to lockably engage with the fluid reservoir.

[0006] The liquid transfer device can be configured to lockably engage with the reaction chamber, e.g., without dispensing, prior to dispensing, and/or after dispensing a liquid sample.

[0007] In some embodiments, the reaction chamber includes one or more components of a biological reaction.

[0008] In another aspect, the disclosure features a liquid transfer device that includes a housing having a pipette tip; and a plunger assembly disposed within the housing and the pipette tip, wherein a portion of the plunger assembly is configured to engage a fluid reservoir such that the plunger

assembly remains stationary relative to the fluid reservoir and the housing moves relative to the plunger assembly.

[0009] In some embodiments, movement of the housing relative to the plunger assembly results in creation of a vacuum within the pipette tip and, optionally, the plunger assembly can be configured to lock in a position resulting in creation of the vacuum. The housing can be configured to move relative to the plunger assembly by pushing the housing down on the fluid reservoir. The device can further be configured to provide an auditory and/or visual indication that the plunger assembly is in a position resulting in the creation of the vacuum.

[0010] A system can include the liquid transfer device and one or more of a fluid reservoir and reaction chamber. When a reaction chamber is included, the reaction chamber can be configured to unlock the plunger assembly when the liquid transfer device and the reaction chamber are interfaced.

[0011] In another aspect, the disclosure features a liquid transfer device configured to draw a sample from a fluid reservoir by pushing the device against the reservoir and systems that include the liquid transfer device and one or both of a reaction chamber and fluid reservoir.

[0012] In the systems described above, two or all three of the liquid transfer device, reaction chamber, and fluid reservoir can have compatible asymmetric cross-sections.

[0013] In another aspect, the disclosure features methods that include (i) obtaining a liquid sample from a sample reservoir using a liquid transfer device described above; and (ii) dispensing the liquid sample, e.g., into a reaction chamber comprising one or more components of a reaction.

[0014] In another aspect, the disclosure features methods that include (i) obtaining a liquid sample from a fluid reservoir using a liquid transfer device (e.g., a liquid transfer device described above); and (ii) dispensing the liquid sample into a reaction chamber, wherein the liquid transfer device sealably engages with the reaction chamber during or prior to dispensing.

[0015] In another aspect, the disclosure features methods that include (i) obtaining a liquid sample from a fluid reservoir using a liquid transfer device (e.g., a liquid transfer device described above); and (ii) dispensing the liquid sample into a reaction chamber, wherein the liquid transfer device lockably engages with the reaction chamber during or prior to dispensing. The methods can further include (iii) interfacing the reaction chamber and the fluid reservoir, such that the reaction chamber lockably engages with the fluid reservoir.

[0016] The systems, apparatuses, and methods disclosed herein can provide for simple analysis of unprocessed biological specimens. They can be used with minimal scientific and technical knowledge, and any knowledge required may be obtained through simple instruction. They can be used with minimal and limited experience. The systems and apparatuses allow for prepackaging or premeasuring of reagents, such that no special handling, precautions, or storage conditions are required. The operational steps can be either automatically executed or easily controlled, e.g., through the use of auditory and/or visual indicators of operation of the systems and apparatuses.

[0017] The details of one or more embodiments of the invention are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

DESCRIPTION OF DRAWINGS

[0018] FIG. 1 is an exploded view of an exemplary system as described herein.

[0019] FIGS. 2A-2C are exploded views of system sub-assemblies.

[0020] FIG. 2D is a view of the system mated and joined.

[0021] FIGS. 3A-3D depict the system in use.

[0022] FIG. 4 depicts the system in the context of an exemplary detection device.

[0023] FIGS. 5A-5C depict the system in cross-section during sample collection.

[0024] FIGS. 6A-6D depict the system in cross-section during sample dispensing.

[0025] FIGS. 7A-7B depict single (7A) and double (7B) variants of the system.

DETAILED DESCRIPTION

[0026] This application describes systems, apparatuses, and methods for transfer of liquids and processing of biological reactions (e.g., nucleic acid amplification reactions). [0027] Referring to FIG. 1, the system can include three subassemblies: a transfer device 100, amplification chamber 200, and an elution container 300. Each subassembly can have a D-shaped or otherwise asymmetrical cross section 105, 205, 305 that is compatible with the other two subassemblies, such that the subassemblies may only be mated to each other in one orientation.

[0028] FIGS. 2A-2C, show exploded views of the subassemblies 100, 200, and 300, respectively. In FIG. 2A, the transfer device 100 includes a body 110 having a D-shaped or otherwise asymmetrical cross section 105 and a pipette tip 120. The transfer device also includes a plunger unit 130 having a syringe plunger 135 that seals within the pipette tip 120 using an o-ring 140. The plunger unit also includes flexible arms 131 having tabs 138 that are aligned with two sets of lower 112 and upper 113 slots in the body 110. Ridges within the body 110 align with grooves in the plunger unit 130 to guide the plunger unit 130 up and down within the body 110. When the plunger unit 130 is in the lower position, the tabs 138 insert into the lower slots 112. When the plunger unit 130 is in the upper position, the tabs 138 insert into the upper slots 113. A spring 150 fits over a spring guide 139 of the plunger unit 130, and can be compressed against the cap 160 when the transfer device 100 is assembled. When the plunger unit 130 is in the upper position, an indicator 137 at the top of the spring guide 139 is visible through an indicator window 165 in the cap 160.

[0029] In FIG. 2B, the amplification chamber 200 includes a body 210 having a D-shaped or otherwise asymmetrical cross-section 205 that is compatible with the cross-section 105 of the transfer device 100. The amplification chamber body 210 also includes two tabs 215 that insert into either the lower slots 112 or upper slots 113 of the transfer device 100 when the two subassemblies are mated. The reaction chamber 200 also includes a microtube 220 having a retaining ring 225 that holds the microtube 220 within an aperture in the bottom of the amplification chamber body 210. The microtube 220 can also have a seal 228 that covers the mouth 223 of the tube 220. In some embodiments, the microtube 220 is optically permeable to allow monitoring of its contents. The amplification chamber 200 also includes a scaling component 230 that fits within the amplification chamber body 210 and over the microtube 220, holding it in

place. The scaling component 230 includes a pliant gasket 235 configured to seal against the pipette housing 180 when the two subassemblies are mated (see FIGS. 6A-6D). Two side tabs 240 are present near the bottom of the body 210 of the amplification chamber 200.

[0030] In FIG. 2C, the elution container 300 has a D-shaped or otherwise asymmetrical cross-section 305 that is compatible with the cross-section 105 of the transfer device 100. The elution container 300 includes an elution buffer reservoir 310 and a guide ring 320 compatible with a pipette housing 180 of the transfer device 100. A seal can cover the mouth of the buffer reservoir 310 or guide ring 320. Two notches 340 are present on the side walls 350 of the elution chamber 300, into which insert the side tabs 240 of the amplification chamber 200 when the two subassemblies are mated.

[0031] FIG. 2D shows the three subassemblies of the system mated and joined for disposal. The transfer device 100 locks into the amplification chamber 200 by insertion of the amplification chamber tabs 215 into the upper slots 113 of the transfer device 100. Similarly, the amplification chamber 200 locks into the elution chamber 300 by insertion of the side tabs 240 of the amplification chamber 200 into the notches 340 of the elution chamber 300. In this configuration, the patient sample and any amplified nucleic acids are sealed within the system to prevent contamination. Approximate dimensions of the joined system are shown.

[0032] FIGS. 3A-3D show an overview of the system in operation. In FIG. 3A, the transfer device 100 is positioned above the elution chamber 300 with their D-shaped cross-sections 105 and 305 aligned. In FIG. 3B, the transfer device 100 is pushed down on the elution chamber 300, such that the pipette tip 120 enters the buffer reservoir 310 and the plunger unit 130 remains stationary relative to the body 110 due to contact with a guide ring on the buffer reservoir 310. This results in the plunger unit 130 in the upper position, compressing the spring 150 such that the indicator 137 shows through the indicator window 165. The presence of the indicator 137 in the indicator window 165 and an audible click as the tabs 138 insert into the upper slots 113 provide auditory and visual feedback that the transfer device has been manipulated properly such that the pipette tip 120 is able to withdraw a portion of the sample from the buffer reservoir 310. In FIG. 3C, the transfer device 100 has been removed from the elution chamber 300 and positioned above the amplification chamber 200 with their D-shaped cross-sections 105 and 205 aligned. In FIG. 3D, the transfer device 100 is pushed onto the amplification chamber 200. The two tabs 215 of the amplification chamber 200 insert into the upper slots 113 of the transfer device 100, displacing the tabs 138 and allowing the compressed spring 150 to relax and the plunger unit 130 to return to the lower position. The indicator 137 is no longer visible in the indicator window 165, signaling that the contents of the pipette tip 120 have been emptied into the microtube 220. The transfer device 100 is locked into the amplification chamber 200 by insertion of the amplification chamber tabs 215 into the upper slots 113 of the transfer device 100.

[0033] FIG. 4 shows the system with an exemplary detection device 400. The detection device 400 includes a first station 410 adapted to securely hold the elution chamber 300 and a second station 420 adapted to securely hold the amplification chamber 200. When in use, the transfer device 100 is moved between the elution chamber 300 at the first

station 410 and the amplification chamber 200 at the second station 420. The detection device includes a lid 430 that can be closed when the detection device 400 is in operation or for storage. A touchscreen user interface 440 is present for inputting data and displaying information regarding the assay. The second station 420 can include a bar code reader or similar device to automatically detect a bar code or similar code present on the amplification chamber 200. The first 410 and second 420 stations can be adapted to heat or cool the contents of the elution chamber 300 and reaction chamber 200. The second station 420 can also be adapted to provide optical, fluorescence, or other monitoring and/or agitation of the microtube 220.

[0034] FIGS. 5A-5C show the system in cross-section during sample collection. In FIG. 5A, the transfer device 100 is placed above the elution chamber 300 such that their cross sections 105, 305 are aligned. The plunger unit 130 is in the lower position and the tabs 138 are in the lower slots 112. In FIG. 5B, the transfer device 100 is lowered until one or more flanges 139 on the lower surface of the plunger unit 130 contact the guide ring 320, and the pipette tip 120 and plunger tip 132 are inserted into the liquid sample 360. The liquid sample 360 can be a patient or other sample or include a patient or other sample dissolved or suspended in a buffer. In FIG. 5C, the transfer device 100 is pushed down by the user into the elution chamber 300. The plunger unit 130 remains stationary through the contact of the one or more flanges 139 against the guide ring 320, while the transfer device body 110 is lowered relative to the plunger unit 130 and elution chamber 300. Simultaneously, a guide channel 116 in the transfer device is pushed downward relative to the guide ring 320. The downward motion of the transfer device body 110 causes the pipette tip 120 to move downward relative to the plunger tip 132 and draw a liquid sample portion 365 into the pipette tip 120. The downward motion of the transfer device body 110 relative to the plunger unit 130 also compresses the spring 150, moves the tabs 138 from the lower slots 112 to the upper slots 113, and causes the indicator 137 to be visible through the indicator window 165. The transfer device 100 with the liquid sample portion 365 can now be lifted off of the elution chamber 300 and is ready for transfer and dispensing.

[0035] FIGS. 6A-6D show the system in cross-section during sample dispensing. In FIG. 6A, the transfer device 100 is placed above the amplification chamber 200 such that their cross sections 105, 205 are aligned. The amplification chamber 200 is held within the second station 420 of the detection device 400 with the microtube 220 seated within a tube holder 428. In FIG. 6B, the transfer device 100 is lowered until two inner tabs 250 within the amplification chamber 200 engage two ridges 170 in the lower sides of the transfer device body 110, the tabs 215 insert into the lower slots 112 of the transfer device 100, and the gasket 235 engages the pipette housing 180. This prevents the transfer device 100 from being easily removed from the amplification chamber 200 once dispensing has been started and prevents release of the sample. In FIG. 6C, the transfer device 100 is further lowered onto the amplification chamber 200, such that the amplification chamber tabs 215 insert into the upper slots 113 of the transfer device and displace the plunger unit tabs 138. Simultaneously, the pipette tip 120 pierces the seal 228 on the microtube 220. In FIG. 6D, the plunger unit 130, no longer held in the upper position, moves to the lower position as the spring 150 expands. This causes

the plunger tip 132 to move downward within the pipette tip 120 and dispense the liquid sample portion 365 into the microtube 220. The liquid sample portion 365 rehydrates a dried reagent pellet 280 in the microtube 220, initiating reaction (e.g., an amplification reaction). The transfer device 100 is locked in place on the amplification chamber 200 by the tabs 215 inserted into the upper slots 113, and any product of the amplification reaction is sealed within the unit by the gasket 235.

[0036] FIGS. 7A and 7B are three-quarter cross sections showing the system configured for one or two microtubes 220. FIG. 7A shows the transfer device 100 and amplification chamber 200 as described above with one pipette tip 120 and one microtube 220. FIG. 7B shows the transfer device 100 and amplification chamber 200 with two pipette tips 120 and two microtubes 220. Using the device in FIG. 7B, parallel reactions (e.g., amplification reactions) can be performed on two portions of one sample.

[0037] The systems and apparatuses disclosed herein can be used to perform reactions, e.g., utilizing biological components. In some embodiments, the reactions involve production of nucleic acids, such as in nucleic acid amplification reactions. Exemplary nucleic acid amplification reactions suitable for use with the disclosed apparatuses and systems include isothermal nucleic acid amplification reactions, e.g., strand displacement amplification, nicking and extension amplification reaction (NEAR) (see, e.g., US 2009/0081670), and recombinase polymerase amplification (RPA) (see, e.g., U.S. Pat. Nos. 7,270,981; 7,666,598). In some embodiments, a microtube can contain one or more reagents or biological components, e.g., in dried form (see, e.g., WO 2010/141940), for carrying out a reaction.

[0038] The systems and apparatuses disclosed herein can be used to process various samples in reactions, e.g., utilizing biological components. In some embodiments, the samples can include biological samples, patient samples, veterinary samples, or environmental samples. The reaction can be used to detect or monitor the existence or quantity of a specific target in the sample. In some embodiments, a portion of the sample is transferred using a transfer device as disclosed herein.

[0039] In some embodiments, a liquid transfer device or pipette tip disclosed herein can be configured to collect and dispense a volume between 1 μ l and 5 ml (e.g., between any two of 1 μ l, 2 μ l, 5 μ l, 10 μ l, 20 μ l, 50 μ l, 100 μ l, 200 μ l, 500 μ l, 1 ml, 2 ml, and 5 ml).

[0040] The disclosure also features articles of manufacture (e.g., kits) that include one or more systems or apparatuses disclosed herein and one or more reagents for carrying out a reaction (e.g., a nucleic acid amplification reaction).

[0041] A number of embodiments of the invention have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the invention. For example, a transfer device as described herein can include three or more pipette tips. Accordingly, other embodiments are within the scope of the following claims.

What is claimed is:

1. A liquid transfer system comprising:
 - a liquid transfer device comprising a housing and at least one pipette tip;
 - a reaction chamber configured to sealably engage with the liquid transfer device; and

a fluid reservoir configured to engage with at least one of the liquid transfer device or the reaction chamber, wherein at least two of the liquid transfer device, reaction chamber, and fluid reservoir have compatible asymmetric cross-sections.

2. The system of claim 1, wherein the liquid transfer device further comprises a plunger assembly disposed within the housing.

3. The system of claim 2, wherein the plunger assembly is configured to remain stationary relative to the fluid reservoir when the liquid transfer device engages with the fluid reservoir.

4. The system of claim 1, wherein the liquid transfer device is configured to lockably engage with the reaction chamber.

5. The system of claim 1, wherein the reaction chamber is configured to lockably engage with the fluid reservoir.

6. The system of claim 1, wherein the reaction chamber contains one or more components of a biological reaction.

7. The system of claim 1, wherein the liquid transfer device is configured to provide an auditory or visual indication when a sample has been drawn into the at least one pipette tip.

8. A method of transferring a liquid sample, comprising: obtaining a liquid sample from a fluid reservoir using a liquid transfer device; engaging the liquid transfer device with a reaction chamber; and dispensing the liquid sample into the reaction chamber, wherein at least two of the liquid transfer device, reaction chamber, and fluid reservoir have compatible asymmetric cross-sections.

9. The method of claim 8, further comprising lockably engaging the liquid transfer device with the reaction chamber prior to or during dispensing.

10. The method of claim 8, further comprising interfacing the reaction chamber with the fluid reservoir after dispensing the liquid sample.

11. A liquid transfer device comprising: a housing having at least one pipette tip; a plunger assembly disposed within the housing; and an asymmetric cross-section configured to mate with a complementary asymmetric cross-section of a separate component.

12. The liquid transfer device of claim 11, wherein the plunger assembly is configured to lock in a position that creates a vacuum within the at least one pipette tip.

13. The liquid transfer device of claim 11, further comprising an indicator configured to provide a visual or auditory signal when a sample has been drawn into the at least one pipette tip.

14. A system for performing a biological reaction, comprising:

a liquid transfer device;
a reaction chamber containing one or more biological reagents; and
a fluid reservoir,
wherein the system is configured to allow transfer of a sample from the fluid reservoir to the reaction chamber using the liquid transfer device, and
wherein at least two components of the system have compatible asymmetric cross-sections to ensure proper alignment during use.

15. The system of claim 14, wherein the reaction chamber is configured to initiate a biological reaction upon receiving the sample from the liquid transfer device.

16. The system of claim 1, wherein the at least one pipette tip comprises a plurality of pipette tips.

17. The system of claim 16, wherein the plunger assembly comprises a plurality of syringe plungers, each syringe plunger operably connected to a corresponding pipette tip of the plurality of pipette tips.

18. The system of claim 1, wherein the fluid reservoir comprises an inner wall and an outer wall, and wherein the liquid transfer device is configured to engage with the fluid reservoir between the inner wall and the outer wall.

19. A method of preparing a biological reaction, comprising:

obtaining a sample using a liquid transfer device from a fluid reservoir;
transferring the sample to a reaction chamber containing dried biological reagents;
sealing the reaction chamber with the liquid transfer device; and
initiating a biological reaction within the sealed reaction chamber.

20. The method of claim 19, further comprising detecting a product of the biological reaction while the reaction chamber remains sealed by the liquid transfer device.

* * * * *